

Carlos Gael Hernandez Castro

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EDUCATION

University of Texas at El Paso (UTEP)

El Paso, TX

Bachelor of Science in Electrical Engineering, Minor in Finance. GPA: 4.0/4.0

Expected Graduation: May 2029

Honors: College of Engineering Honors Program; Hispanic Scholarship Fund Scholar; UTEP Excellence Scholarship

Relevant Coursework: Electromagnetism with Lab (in progress); self-study in digital logic and computer architecture

EXPERIENCE

Ground Station Engineer

Jan 2025 – Present

Sun City Rocket Team (UTEP)

El Paso, TX

- Designed a real-time telemetry ground station that decodes framed DAQ packets and plots pressure, temperature, and inertial data in flight; supported 2nd place at IREC 2025 (30k ft COTS) and 1st at the 2025 Lone Star Cup.
- Built an Arduino Mega DAQ shield that conditions 4-20 mA pressure transducers (precision shunt, op-amp front end) and K-type thermocouples, with ADC scaling and packet framing so the ground station receives calibrated values.

Hardware and AI Data Intern

Jun 2026 – Aug 2026

ArcellAI, Physical AI and robotics data platform

San Francisco, CA

- Built a standalone Python data loader for LIBERO, a robot-learning benchmark, adapting PyTDC's inheritance-based loader architecture and classifying grasp-prediction tasks as single- vs. multi-instance prediction.
- Managed GitHub repository access and structure, onboarded new team members, and tracked sprints in ClickUp.

FIRE Venture Fellow

Mar 2026 – Jun 2026

Hunt Center for Entrepreneurship, NMSU Arrowhead Center

Las Cruces, NM

- Evaluated a live regional startup on market, team, traction, and cap table (seed and Series A dilution, vesting), and wrote an investment thesis on AI-powered grid orchestration (FERC Order 2222).

Undergraduate Researcher

Sep 2025 – Nov 2025

Miner Spark Research Project, University of Texas at El Paso

El Paso, TX

- Developed a modular UAV attachment that runs facial verification on-device (edge computer vision) for low-cost drones.
- Measured recognition accuracy across lighting levels (lux) to validate the system's hardware-software integration.

PROJECTS

The Sense Cube: Phone-to-Braille Reader | Founder and Hardware Lead

Ongoing

- Designed a 2-layer KiCad evaluation board that drives 48 solenoid dots across 8 Braille cells from a few Arduino pins, using four 74HC238 decoders with TBD62783 source and TD62083 Darlington sink drivers for per-dot control.
- Designed the 7.6 × 29.7 mm 2-layer Braille cell PCB with split source/sink headers, a third-revision Fusion 360 solenoid carrier, and a fixture for winding 50 µm magnet-wire electromagnets.
- Won \$6,000 across three pitch competitions, including 1st of 60+ entries at UTEP Miner Pitch; 2nd at Miner Tank.

Transistor-Level Digital Logic | Independent

- 4-bit calculator:** Built a 4-bit adder/subtractor from 200+ discrete BJTs with no logic ICs or microcontroller, from a NAND-derived gate set (NOT, AND, OR, XOR) up to chained full adders and a transistor-level 7-segment decoder.
- Up/down counter:** Designed a 000-999 counter/timer with a NAND relaxation-oscillator clock (~3.5 kHz), AND-gate decade detection that forces BCD rollover and carry, and up/down, reset, and pause control.

Wayve Smart Navigation Cane | BorderHack '26, 1st Place

2026

- Built in 24 hours with a 4-person team: a cane that turns Google Maps Directions API steps into wrist cues from 3 handle servos, with ultrasonic obstacle sensing that ramps vibration inside ~1.5 m.

AWARDS & LEADERSHIP

- SOLACYT International Competition:** Top 3 of 25,000+ students across Ibero-America (Chile, Colombia, Uruguay, Mexico) with a robotic skeleton puppet that improved students' anatomy knowledge by ~60%.
- Robotics Team Lead (2022):** Led a 6-person team that raised \$750+ and built a custom electric wheelchair; founded a robotics workshop for 20+ students.
- Dell Tech Academy Case Competition:** 3rd Place in Dell Technologies' two-day global supply chain design challenge.

TECHNICAL SKILLS

Hardware: Circuit design, schematic capture, 2-layer PCB layout, analog signal conditioning (op-amps, 4-20 mA, thermocouples), solenoid driver stages, transistor-level digital logic, AC-DC rectifier supplies, DAQ, soldering, prototyping

Programming and Embedded: Python, C, C++, MATLAB; Arduino, ESP32, Raspberry Pi, Bluetooth

Tools: KiCad, Fusion 360 (Autodesk Certified User), Git/GitHub, FDM 3D printing | **Languages:** English, Spanish (fluent)